TEXORA · LECTURE SHEET

Vector Calculus

Vector fields, gradient, divergence and curl with their physical meanings — then line, surface and volume integrals tied together by the theorems of Green, Gauss and Stokes.

MATH-1201 · MATHEMATICS-II · TOPIC 1 · LEVEL 1 TERM 2

1. Scalar and vector fields

A **scalar field** assigns a number to each point of a region: temperature T(x, y, z), density, pressure, potential. A **vector field** assigns a vector: fluid velocity v(x, y, z), force, electric field. Everything in this topic is calculus done on such fields. The operator that organises it all is **del** (nabla):

$$\nabla = \mathrm{i}\,\partial/\partial x + \mathrm{j}\,\partial/\partial y + \mathrm{k}\,\partial/\partial z$$

Applied three ways to the fields it produces the gradient (of a scalar), the divergence (of a vector) and the curl (of a vector).

2. Gradient — direction of steepest ascent

For a scalar field φ, the **gradient** is the vector

$$\nabla\varphi = (\partial\varphi/\partial x)\,\mathrm{i} + (\partial\varphi/\partial y)\,\mathrm{j} + (\partial\varphi/\partial z)\,\mathrm{k}$$

Its meaning: ∇φ points in the direction in which φ increases fastest, and |∇φ| is that fastest rate. It is always **normal (perpendicular) to the level surfaces** φ = constant. The rate of change in any unit direction û (the **directional derivative**) is the component ∇φ · û.

WORKED EXAMPLE

φ = x²yz. ∇φ = (2xyz, x²z, x²y). At (1, 1, 1): ∇φ = (2, 1, 1).
Directional derivative along (1, 2, 2): û = (1, 2, 2)/3, so $D_{\hat{u}}\varphi = (2\cdot1 + 1\cdot2 + 1\cdot2)/3 = 6/3 = \mathbf{2}$.

For the sphere field φ = x² + y² + z²: ∇φ = (2x, 2y, 2z) — pointing radially outward, normal to the spherical level surfaces, exactly as the meaning demands.

3. Divergence — outflow per unit volume

For a vector field $F = F_1\mathrm{i} + F_2\mathrm{j} + F_3\mathrm{k}$, the **divergence** is the scalar

$$\text{div } F = \nabla \cdot F = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$$

Physical meaning: for a velocity field, div F at a point is the net rate of outflow of fluid per unit volume there. div F > 0 marks a **source**, div F < 0 a **sink**, and div F = 0 an **incompressible (solenoidal)** flow — what flows in flows out. In electromagnetism, div E = ρ/ε_0 says charge is the source of the electric field.

WORKED EXAMPLE

$F = x^2 i + y^2 j + z^2 k$: div F = 2x + 2y + 2z; at (1, 1, 1) div F = 6 — a source point.

F = −yi + xj (rigid rotation): div F = 0 + 0 = 0 — rotation circulates fluid without net outflow anywhere.

4. Curl — rotation per unit area

The **curl** is the vector

$$\text{curl } F = \nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Physical meaning: curl F at a point measures the local rotation (circulation density) of the field; its direction is the axis about which the rotation is greatest. A tiny paddle wheel placed in a fluid spins about the curl's direction with speed proportional to |curl F|. A field with curl F = 0 everywhere is **irrotational** — and then F is the gradient of a potential (conservative field).

WORKED EXAMPLE

F = −yi + xj: curl F = (0 − 0)i + (0 − 0)j + (1 − (−1))k = **2k** — uniform rotation about the z-axis, matching the rigid-rotation picture (angular velocity ω = 1, curl = 2ω).

F = ∇φ for any φ always has curl zero: rotation cannot be manufactured from a potential.

5. Line integrals and work

The **line integral** of F along a curve C is $\int_C F \cdot dr$ — physically, the **work** done by a force field along a path. It is computed by parametrising C: r(t), substituting, and integrating an ordinary integral in t. If F is conservative (F = ∇φ), the integral depends only on the endpoints: ∫ = φ(end) − φ(start), and round any closed loop it is zero.

WORKED EXAMPLE

F = (x + y)i + (x − y)j along the line y = x from (0, 0) to (1, 1). Put x = t, y = t, dr = (i + j) dt: F = 2t i + 0 j, so F · dr = 2t dt, and work = $\int_0^1 2t\,dt$ = **1**.

6. Surface integrals and flux

The **flux** of F across a surface S is $\iint_S F \cdot \hat{n}\, dS$ — the volume of fluid crossing S per unit time when F is a velocity (times density for mass flux). For a parametric or projected surface, $F \cdot \hat{n}\, dS$ is rewritten in the surface's own variables; for a plane region the projection formula $dS = dx\, dy/|\hat{n} \cdot k|$ is the workhorse. Flux across a **closed** surface, taken outward, is the bridge to the divergence theorem.

7. The three great theorems — one pattern

THEOREM	STATEMENT	MEANING
Green (plane)	$\oint_C (P\,dx + Q\,dy) = \iint_R (\partial Q/\partial x - \partial P/\partial y)\, dA$	Circulation round a plane region = the curl (z-component) summed over its interior.
Gauss (divergence)	$\iint_S F \cdot \hat{n}\, dS = \iiint_V \text{div } F\, dV$	Total outflow through a closed surface = the sources summed over the volume inside.
Stokes	$\oint_C F \cdot dr = \iint_S (\text{curl } F) \cdot \hat{n}\, dS$	Circulation round a loop = the curl flux through any surface spanning the loop.

WORKED VERIFICATIONS

Gauss: F = xi + yj + zk over the unit sphere. div F = 3; volume = 4π/3; flux = 3 × 4π/3 = **4π** — and indeed at the surface $F \cdot \hat{n} = 1$ everywhere on a sphere of area 4π.
Stokes: F = −yi + xj round the unit circle in z = 0. curl F = 2k; flux through the disc = 2 × π(1)² = 2π. Directly: x = cosθ, y = sinθ gives $F \cdot dr = (\sin^2\theta + \cos^2\theta)\, d\theta = d\theta$, so $\oint = \int_0^{2\pi} d\theta$ = **2π** ✓.
Green: P = xy, Q = x² over the triangle (0,0), (1,0), (0,1): $\partial Q/\partial x - \partial P/\partial y = 2x - x = x$; $\iint x\, dA = \bar{x}$ × Area = (1/3)(1/2) = **1/6**, the value the three-segment line integral also returns.

8. Key points

- ∇φ: steepest ascent, normal to level surfaces; directional derivative = ∇φ · û.
- div F: net outflow per unit volume; zero = solenoidal/incompressible.
- curl F: local rotation; zero = irrotational = conservative = path-independent line integrals.
- Identities worth knowing: div(curl F) = 0 and curl(∇φ) = 0, always.
- Green, Gauss and Stokes are one idea in three dimensions of dress: the integral of a derivative over a region equals the field's integral over the region's boundary.

9. Practice

Answer in your own words; brief model answers follow.

1. Find $\nabla\varphi$ at (1, 2, 3) for $\varphi = x^3 + y^3 + z^3 - 3xyz$.
2. For $F = (x^2 - y^2)i + 2xy\,j$, compute div F and curl F at (1, 1), and comment on the pair of values.
3. Show that $F = (2xy + z)i + x^2 j + xk$ is conservative by finding its potential φ, and hence evaluate $\int_C F \cdot dr$ from (0, 0, 0) to (1, 1, 1) along any path.
4. Use the divergence theorem to find the flux of $F = x^2 i + y^2 j + z^2 k$ outward through the unit cube $[0, 1]^3$.
5. A fluid has velocity $v = (2x, -y, -z)$. Is the flow compressible at the origin? Where is div v zero?
6. State Green's theorem and use it to evaluate $\oint_C (y^2\,dx + x^2\,dy)$ round the unit circle.

Answer key

1. $\nabla\varphi = (3x^2 - 3yz, 3y^2 - 3zx, 3z^2 - 3xy)$; at (1, 2, 3): $(3 - 18, 12 - 9, 27 - 6) = (-15, 3, 21)$.
2. div $F = 2x + 2x = 4x \to 4$ at (1, 1). curl $F = (\partial(2xy)/\partial x - \partial(x^2 - y^2)/\partial y)\,k = (2y + 2y)\,k = 4y\,k \to 4k$ at (1, 1). The field both diverges and rotates at that point — divergence and curl are independent properties; neither implies the other.
3. $\partial\varphi/\partial x = 2xy + z \to \varphi = x^2y + zx + g(y, z)$; $\partial\varphi/\partial y = x^2$ ✓ with g independent of y; $\partial\varphi/\partial z = x + \partial g/\partial z$ must equal x, so g is constant: $\varphi = x^2y + zx$. Since $F = \nabla\varphi$, the integral is path-independent: $\varphi(1,1,1) - \varphi(0,0,0) = (1 + 1) - 0 = 2$.
4. div $F = 2x + 2y + 2z$. $\iiint$ over the unit cube $= 2(\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2})\ldots$ precisely $\int_0^1 2x\,dx = 1$ for each of the three terms, total flux = 3.
5. div $v = 2 - 1 - 1 = 0$ everywhere, including the origin: the flow is incompressible (solenoidal) at every point — the x-direction expansion is exactly fed by the y- and z-direction contractions.
6. $\oint (P\,dx + Q\,dy) = \iint (\partial Q/\partial x - \partial P/\partial y)\,dA$. Here $\partial Q/\partial x - \partial P/\partial y = 2x - 2y$; over the unit disc both $\iint 2x\,dA$ and $\iint 2y\,dA$ vanish by symmetry, so the circulation is 0.
